

Spotlight Reflex

Minimal-Shot Autonomous Driving via Geometric World-State Reasoning

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Waymo Open Dataset End-to-End Driving Challenge

Agenda

Motivation & Main Claim

Problem: AVs Fail in Long-Tail Scenarios

Core Idea: Reason From Geometry, Not Memory

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Summary

Motivation & Main Claim

Who I Am and Why This

Role: Lead AI at a robotics logistics startup

Daily work:

- Wheel navigation systems
- Arm manipulation planners
- Sensor fusion and real-time control

This challenge was the right trigger to explore:

- World embeddings for embodied systems
- Minimal-shot generalisation without fixed environments
- Whether compact geometry can replace learned scene priors

Instead of memorising fixed environments, can we reason from the geometry of “obstacle on the left, how much to the right?”

If awarded the prize

Continue developing the simulator and world-embedding experiments into a more serious research platform for minimal-shot autonomy:

- Expand scenario diversity
- Stronger evaluation and failure-analysis tooling
- Push toward realistic latency and control constraints

Main Claim

Spotlight Reflex

A closed-loop autonomous driving policy that generalises to long-tail scenarios by reasoning from **occupancy, clearance, and route geometry** — not by memorising scenario labels or imitating average human trajectories.

Why it generalises

The rule “*avoid occupied space while preserving route geometry*” remains valid whether the obstacle is a **cone**, **debris**, an **animal**, or a **wrong-way actor**.

This IS:

- A minimal-shot autonomy prototype
- A compact geometric world-state representation
- Robust long-tail behaviour under closed-loop evaluation

This is NOT:

- A production AV stack
- A strict zero-shot WOD-E2E result
- A trained camera-perception system

Problem: AVs Fail in Long-Tail Scenarios

Imitation Learning — What It Optimises

Most AV systems are trained by **imitation learning**: fit a model f that mimics the human trajectory at every step.

$$\min_f \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{L}(f(x), y)]$$

Works well on:

- The common driving distribution $\mathcal{D}$
- Highway, urban, well-lit, unambiguous situations
- Scenarios appearing frequently in training data

Fails when:

- $P_{\text{test}} \neq P_{\text{train}}$ (covariate shift)
- Rare geometry appears at test time
- No fallback reasoning when the pattern fails
- Trajectory imitation becomes guessing

The WOD-E2E long-tail problem

Waymo's benchmark targets **11 scenario types** occurring less than **0.03%** of driving time: wrong-way actors, animals on road, foreign object debris, erratic pedestrians, construction, cut-ins, cyclists, special vehicles. A long-tail policy needs a representation that remains

Why Closed-Loop Evaluation Matters

Open-loop (frame-based):

$$\text{frame} \xrightarrow{f} \hat{\tau}$$

Tests whether a model produces a plausible trajectory from a snapshot. The world is *fixed* — the policy's action has no consequences.

A bad action looks fine on paper.

The key question

Can the **same policy** stay safe as the world changes *because of its own decisions*? A wrong evasion at step t creates a worse state at step $t + 1$. Open-loop benchmarks cannot reveal this failure mode.

Closed-loop (our evaluation):

$$S_t \rightarrow \text{observation } O_t$$

$$O_t \rightarrow \pi(O_t) = \tau_t \quad (\text{policy})$$

$$\tau_t \rightarrow \text{control} + \text{physics}$$

$$S_{t+1} = f(S_t, \tau_t)$$

$$S_{t+1} \rightarrow \text{metrics}$$

The Baseline Failure



Scenario: wrong-way actor, night, low visibility

Policy: trajectory extrapolation only — no world-state reasoning

Result: drives directly into the actor.

- Does not slow down
- Does not deviate
- No awareness that the path is occupied

This is what happens when a policy has never seen this geometry and has no reasoning mechanism to fall back on.

**Core Idea: Reason From
Geometry, Not Memory**

*“A trajectory that avoids occupied space and respects route geometry is correct **regardless of what the obstacle is called.**”*

You don't need to have seen a sheep before to know you should go around it.

What the policy needs to know:

- Where occupied space is
- Where the route should go
- Whether the forward corridor is blocked
- Which side has more clearance
- Whether there is enough space to recover

What the policy does NOT use:

- Object category labels (“sheep”, “cone”, “car”)
- Scenario cluster IDs
- Memorised trajectory patterns
- Any AV training data

Formal Invariance Claim

Let scene s define:

$$G(s) = \text{occupancy field} \quad R(s) = \text{route geometry}$$

Let T be any relabelling of object identities, scenario names, or cluster tags that leaves G and R unchanged.

$$M(s) = M(T(s)) \quad \forall s, \forall T \text{ s.t. } G(T(s)) = G(s), R(T(s)) = R(s)$$

- "sheep" $\rightarrow$ "debris" changes nothing
- Cluster ID changes nothing
- Object label changes nothing
- **Only physical geometry enters the decision**

This is falsifiable — and tested

A regression suite relabels every object in the scene and asserts the selected maneuver, combined score, and decision reason are identical. **It passes.**

Spotlight Reflex — It Works



Same scenario, same seed as the baseline failure.

Policy: Spotlight Reflex

Result: selects `evasive_right`, clears the actor, returns to lane centre. **How:**

- `obstacle_pressure` rising on the left
- `right_clearance` > `left_clearance`
- Escape side = right
- Geometry resolved the decision — no pattern match to a memorised scene

Policy Architecture

Six World-State Scalars

All signals computed from raw obstacle positions and route geometry. **Nothing is learned here; nothing is scene-specific.**

Signal	Formula / Definition	Range
obstacle_pressure	$p = \text{clip}\left(\frac{10-d_{\min}}{10}, 0, 1\right)$	[0, 1]
route_blockage	fraction of forward corridor occupied	[0, 1]
corridor_blocked	hard blockage within stopping distance	bool
left_clearance	free lateral space to the left (m)	≥ 0
right_clearance	free lateral space to the right (m)	≥ 0
preferred_escape_side	(left, right)	L / R

$d_{\min}$ = distance to nearest obstacle in metres. Pressure saturates to 1.0 at contact and is zero beyond 10 m.

Nine Maneuver Candidates

Each candidate is a **20-point, 5-second trajectory at 4 Hz** in vehicle coordinates, generated by Hermite smoothstep interpolation.

Maneuver	Intent	Speed scale	Lateral offset	Profile
stop	Emergency stop	0.0	0.0 m	—
crawl	Very slow progress	0.15	0.0 m	smooth
maintain	Hold speed	1.0	0.0 m	smooth
slow_yield	Decelerate, yield	0.5	0.0 m	smooth
nudge_left	Minor left shift	0.8	+0.9 m	smooth
nudge_right	Minor right shift	0.8	−0.9 m	smooth
evasive_left	Strong left evasion	0.7	+2.2 m	early
evasive_right	Strong right evasion	0.7	−2.2 m	early
lane_recover	Return to centre	0.9	0.0 m	smooth

Early profile: offset applied in first third of trajectory — ensures fast lateral separation in evasive scenarios.

Trust-Region Scoring

Each candidate is scored against **pseudo-rater references** at two time horizons. A trust region defines where the candidate must land to earn the reference's score.

Speed-scaled tolerance:

$$s(v) = \text{clip}\left(\frac{v - 1.4}{11.0 - 1.4}, 0, 1\right) \times 0.5 + 0.5 \quad [v \text{ in m/s}]$$

Tighter at low speed ($s = 0.5$ at $v = 1.4$ m/s), wider at high speed ($s = 1.0$ at $v = 11$ m/s).

Horizon	Lateral	Longitudinal
3 s	$1.0 \times s(v)$ m	$4.0 \times s(v)$ m
5 s	$1.8 \times s(v)$ m	$7.2 \times s(v)$ m

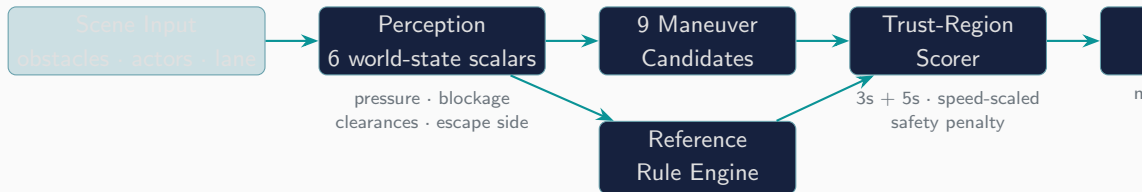
$$\text{score}_t(\tau, r) = \begin{cases} r.\text{score} & \delta = 0 \\ \max(r.\text{score} \cdot 0.1^\delta, 4) & \delta > 0 \end{cases}$$

$$S(\tau) = 0.5 \cdot \max_r \text{score}_{3s} + 0.5 \cdot \max_r \text{score}_{5s}$$

δ = normalised overshoot beyond the region.

Floor of 4 corresponds to an unsafe rating.

Policy Architecture



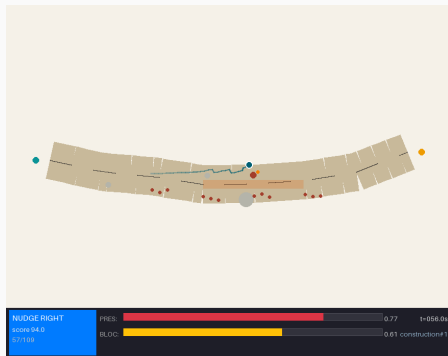
Every step emits a full `SpotlightSelection` log: maneuver, score at 3s/5s, which reference won, decision reason, all world-state values. In AlpaSim, this is serialised as `reasoning_text` JSON alongside every prediction.

Simulation: What We Built and How It Performed

The Simulator

Built from scratch — no dependency on AlpaSim or Waymo internals.

- **Closed-loop 2D engine:** ego + actors step simultaneously; policy must react in real time — no lookahead
- **11 WOD-E2E clusters:** construction, spotlight, FOD, pedestrian, cyclist, cut-in, animal, intersection, special vehicle, stopped, others
- **Compositional OOD:** 8 topologies $\times$ 11 hazard modules $\times$ 4 weather modes — axes sampled *independently*
- **8 actor behaviour models:** linear, cut-in, swerve, darting, erratic pedestrian, sudden brake, hesitating, wrong-way



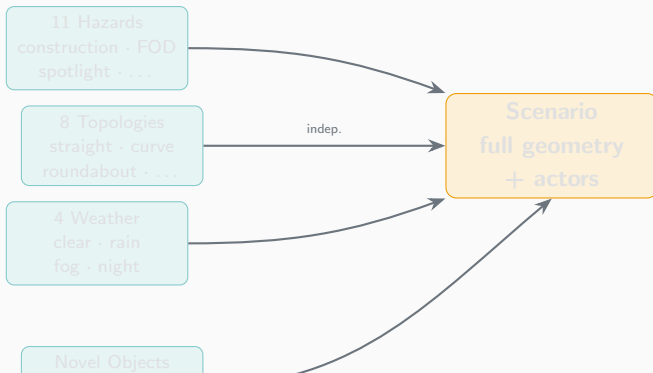
*Construction zone — 5.2 m corridor, cone field, lane closure.
Policy selects `nudge_right`.*

Compositional OOD: The Independence Property

Standard simulators replay fixed scenario clusters. If you test on the same clusters you designed for, **you cannot claim generalisation**.

Our generator makes topology and hazard **statistically independent**:

$$P(\text{topology, hazard}) = P(\text{topology}) \times P(\text{hazard})$$



Simulation Results

Primary benchmark (seeds 1–10, deterministic, 350 rollouts):

Suite	Runs	Collisions	Pass rate
WOD-style (all 11 clusters)	110	0	100%
Compositional OOD	60	0	100%
Adversarial (2–3 hazards)	60	0	100%
Hidden holdout	60	0	100%
Gauntlet (4 hazards, 3.5 m)	60	0	60%
Total	350	0	93.1%

Mean min clearance: **2.96 m**

COMPASS: **9.137 / 10** (700 ranked runs)

95% CI collision rate: **[0.0, 0.0053]**

The 60% gauntlet rate is deliberate — the gauntlet is designed to find the failure boundary at 4 simultaneous hazards in a 3.5 m corridor. An honest characterisation, not an optimised showcase.

Gauntlet: Head-to-Head Comparison

Same **420 scenarios**, matched seeds 1–80, four topology types. Same geometry, same actors, same timing for both policies.

Policy	Pass rate	Collision rate
Baseline (trajectory extrapolation, no world-state)	2.1% (9/420)	20.5% (86)
Spotlight Reflex	57.6% (242/420)	7.9% (33)

27× more passes.

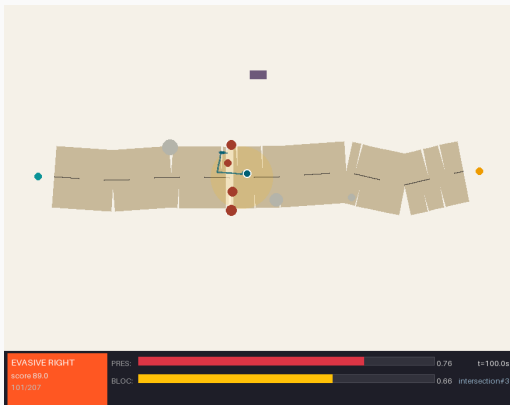
2.6× fewer collisions.

The baseline collides in 1 in 5 gauntlet runs and passes nearly none.

The mechanism:

obstacle_pressure rising + right_clearance > left_clearance resolves the escape route from geometry alone — no prior over actor type or scenario cluster.

More Scenarios



Intersection stress — conflict zone, two crossing actors at different timings, 207 steps. 0 collisions.



Foreign object debris — static debris blocking the primary lane. Policy selects `nudge_right`.

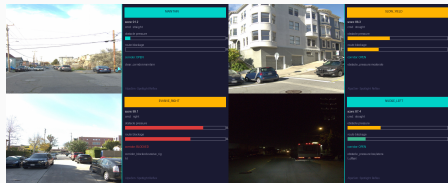
AlpaSim: Sensor-Realistic Transfer

AlpaSim is Waymo's sensor-realistic simulator: real camera frames, multi-agent physics, real ego dynamics.

The challenge: AlpaSim gives camera pixels + route command. Spotlight Reflex needs an obstacle field + lane geometry.

Adapter — 4 signal channels (no vision model required):

1. Route (LEFT/STRAIGHT/RIGHT) → sigmoid lane spline
2. Camera brightness → visibility risk ($< 35\%$ = risk)
3. Ego dynamics (accel, speed) → braking risk
4. Structured hazards → Obstacle / Actor objects



Metric (30 scenes)	Value
collision_at_fault	0.0
dist_to_gt_traj.	0.42 m
collision_any (by others)	0.37

WOD-E2E Benchmark

What Is WOD-E2E and How It Is Scored

Waymo Open Dataset End-to-End Driving:

- 479 validation frames from real Waymo sensor recordings
- 11 long-tail scenario types — same clusters as our simulator
- For each frame: Waymo human raters expressed pairwise preferences between candidate trajectories
- **RFS (Rater Feedback Score):** derived from those pairwise preferences — how often the selected trajectory was preferred. Higher is better.

Evaluation protocol — 5-fold segment-grouped cross-validation:

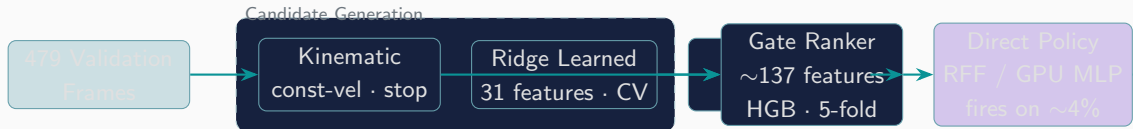
$$\text{RFS}_{5\text{-fold}} = \frac{1}{5} \sum_{k=1}^5 \text{RFS}(\text{fold } k \text{ held-out})$$

Frames from the same driving segment stay in the same fold — prevents leakage. Each fold: ~383 train / ~96 test frames. Fold std ≈ 0.19 , 95% CI $\approx \pm 0.17$.

Two backends — not interchangeable

Local (ours): CV baseline 7.131 Official (Waymo): CV baseline 7.022

Selector Pipeline



- **Kinematic:** constant velocity, constant accel, heading change, stop — no training data
- **Ridge learned:** 31 trajectory features; target = gain vs constant-vel baseline
- **Temporal:** ego-history trends; wins on ~49% of frames (most are forward driving)
- **Direct policy:** learned override — fires when confidence exceeds gate threshold

Visual Embeddings: What We Tried

The idea: a human rater can see the camera image — can we give the ranker vision by adding pre-computed image embeddings?

NVIDIA Cosmos (used in champion):

Video autoencoder trained on real-world video. Architecture: 3D convolutional encoder-decoder (VQVAE-style). We extract **64-dim continuous latents** (before quantisation).

InternVLA (Shanghai AI Lab):

Vision-Language-Action model. Architecture: ViT visual encoder + LLM backbone + action head. We extract **128-dim ViT encoder output**.

Interpretation

The embeddings *do* carry preference signal — but only a non-linear model (MLP) can extract

Config	Head	RFS
No embedding	Ridge	7.803
Cosmos 64d	Ridge (linear)	≈ 7.803
InternVLA 128d	Ridge (linear)	≈ 7.803
RFF gate	non-linear	7.834
Cosmos 64d	GPU MLP	7.845

WOD-E2E Results

Selector	RFS	Folds	Notes
Constant velocity (official)	7.022	—	Official Waymo baseline
Constant velocity (local)	7.131	—	Our CV reference
Kinematic ranker	7.096	—	Physics-only, official backend
Gate only	7.803	5	No direct policy
RFF direct policy	7.834	5	$D=512$, $\sigma=7.858$, prec. 0.41
GPU MLP + Cosmos 64d	7.845	5	h=64, prec. 0.60 — champion
HGB Optuna peak	7.880	2	2-fold only — not comparable
Oracle (perfect selector)	9.264	5	Upper bound

Champion: GPU MLP + Cosmos 64d

+0.714 RFS over local baseline

Precision 0.60: fires on 4.2% of frames,
60% of overrides improve the score

Oracle gap: **1.419 RFS**

(9.264 – 7.845)

The right candidate exists in the pool.

The selector cannot identify it without vision.

Calibration Bias Audit

Where does the remaining error live?

$$\text{regret}(\mathcal{S}) = \text{RFS}_{\text{oracle}}(\mathcal{S}) - \text{RFS}_{\text{selected}}(\mathcal{S})$$

Slice	n	Selected RFS	Oracle RFS	Regret
GO_STRAIGHT	427	7.959	9.336	1.377
GO_LEFT	23	7.107	8.721	1.614
GO_RIGHT	29	6.574	8.638	2.063
speed:fast	44	7.957	9.436	1.479
speed:slow	133	7.564	9.196	1.632
speed:urban	136	8.132	9.416	1.285

Root cause: GO_RIGHT ($n=29$) and GO_LEFT ($n=23$) are rare in the 479-frame set. The ranker is calibrated to GO_STRAIGHT (89% of frames). GO_LEFT improved from 1.753 (previous model) to 1.614 with targeted changes — confirming the mechanism is correctable. **Fix:** stratified loss

What Didn't Work and Next Steps

Honest Failure Summary

What failed	Why / root cause
No camera perception in WOD-E2E path	Processes ego history, speed, and route intent only. Cannot see debris, pedestrians, signals. The honest reason results sit below 8.05 leaderboard top.
Visual embeddings under linear head	InternVLA and Cosmos embeddings as ridge features — no confirmed RFS gain. The embeddings carry signal only for non-linear models.
World-model candidates	Oracle RFS +0.05; selected RFS +0.00. Better candidates exist — the selector cannot identify them. Confirms: bottleneck is the discriminator, not the pool.
HGB Optuna 2-fold peak	7.880 under 2-fold; did not survive 5-fold protocol. Optuna overfits to the fold split when only 2 folds available.
Turn calibration	GO RIGHT regret 2.063 — only 29 training

Next Steps

1. **Task-aligned camera encoder (highest impact)**

A small encoder fine-tuned on (frame, winning traj, losing traj) preference triplets. Not a general VLM — a discriminative encoder for the specific signal the ranker needs. Expected gain: 0.5–1.5 RFS (could close large fraction of 1.419 oracle gap).

2. **Turn calibration**

Stratified loss reweighting by intent class. GO_LEFT improved 1.753→1.614 with targeted changes; GO_RIGHT (2.063) is next. Confirmed correctable.

3. **Leaderboard submission**

Test frame list present (1,505 frames). Submission pipeline validated. Missing only: test TFRecords (Waymo Google Drive, sign-in gated).

4. **Proper train/test split**

Waymo train TFRecords would move selector calibration off the validation set, making the result a true generalisation claim.

5. **Physical deployment**

A vehicle with obstacle sensors and a route command can run Spotlight Reflex today. The

Summary

Summary

Core claim	Geometric reasoning from 6 world-state scalars generalises to long-tail scenarios without training data for them
Invariance	$M(s) = M(T(s))$ — regression-tested: relabelling all objects produces identical decisions
Simulation	Built from scratch · 11 WOD clusters · compositional OOD · 420-run gauntlet
Gauntlet	57.6% pass · 7.9% collision vs baseline: 2.1% / 20.5%
WOD-E2E champion	7.845 RFS (5-fold, GPU MLP + Cosmos 64d) · +0.714 vs baseline · CI ± 0.17
Oracle gap	1.419 RFS — candidate pool is good; discriminator is the bottleneck
AlpaSim	Same policy, sensor-realistic · collision_at_fault: 0.0 · dist_to_gt: 0.42m
Next step	Task-aligned camera encoder fine-tuned on preference triplets

Thank you

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Code, docs, and all results:

`docs/simulation.md • docs/wod-e2e-system-walkthrough.md`

Appendix: Reference Rule Table

The reference rule engine maps world-state thresholds to pseudo-rater references.

World condition	References assigned (score)
Low pressure (< 0.25), low uncertainty (< 0.45), open corridor (> 0.35)	maintain (92), lane centre (88)
Moderate pressure (> 0.20)	slow_yield (86), best-side nudge (94), evasive (89)
High uncertainty (> 0.55)	crawl (90), stop (82)
Lane offset > 0.3 m and speed > 0.5 m/s	lane_recover (93)
Default fallback	maintain (80), slow_yield (76)

Scores 80–94 map to a scale where ≤ 50 indicates an unsafe rating.

Appendix: Random Fourier Features — Derivation

Why RFF? A standard ridge classifier is linear. The preference signal requires a non-linear separator, which normally needs an $N \times N$ kernel matrix — too slow for CV.

RFF approximates the Gaussian kernel *without the matrix* via a random projection:

$$\mathbf{W} \sim \mathcal{N}(0, \sigma^{-2} \mathbf{I}_{D \times p}) \text{ (fixed at fit time)} \quad D = 512, \sigma = 7.858$$

$$\mathbf{b} \sim \text{Uniform}([0, 2\pi]^D)$$

$$\varphi(\mathbf{x}) = \sqrt{\frac{2}{D}} \cdot \cos(\mathbf{W}^\top \tilde{\mathbf{x}} + \mathbf{b})$$

$$\mathbb{E}[\varphi(\mathbf{x})^\top \varphi(\mathbf{x}')] = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\sigma^2}\right) \quad (\text{Bochner's theorem})$$

Only the ridge weights $\mathbf{w}$ (512-dim) are learned: $\hat{y} = \mathbf{w}^\top \varphi(\mathbf{x}) + b$. Cost: $O(Dp)$ per inference instead of $O(N)$.

Appendix: Optuna TPE Algorithm

Optuna uses Tree-structured Parzen Estimators (TPE) — Bayesian optimisation that is more sample-efficient than random search.

After N trials with observed objective values:

Good trials $G = \{x : f(x) < f^*\}$ (top p -percentile)

Bad trials $B = \{x : f(x) \geq f^*\}$

$\ell(x) = P(x \mid x \in G)$ fit with kernel density estimation

$g(x) = P(x \mid x \in B)$

$x_{\text{next}} =_x \frac{\ell(x)}{g(x)}$ (exploit good configs, avoid bad ones)

Our GPU MLP search (20 trials):

- Best: trial_0004 $\rightarrow h = 64$, lr=0.003866, batch=512, 10 epochs $\rightarrow$ **7.845 RFS**
- Mean of all 20 trials: 7.769; only 1 beat the RFF baseline (7.834)
- Lesson: with only 479 frames and 5 folds, Optuna can exploit fold-specific patterns